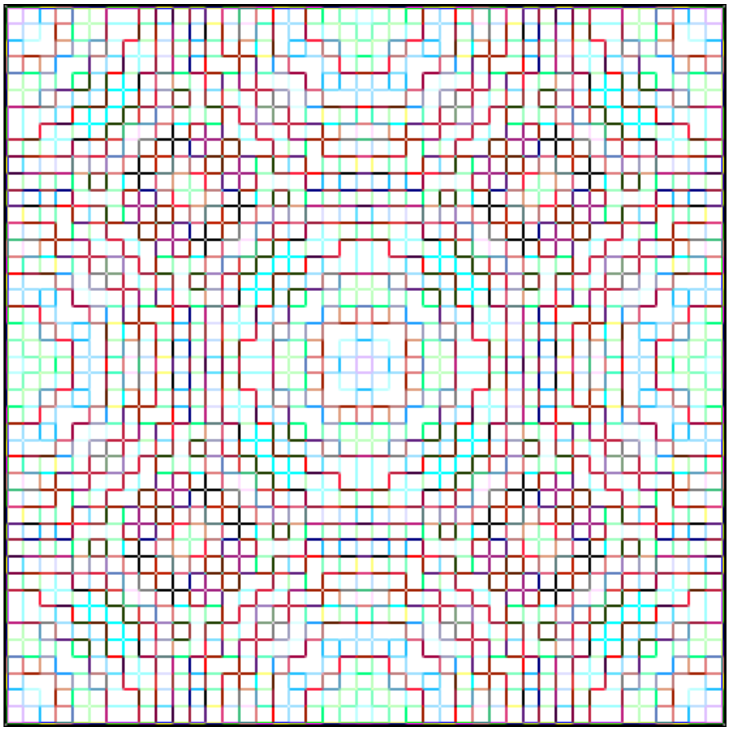
On the Digital Root, the Vedic Square, and the Tiling Properties of Discrete Samplings of Periodic Functions



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Introduction

My friend Kaya (Karel Lavir in full) found an interesting “magic square” about 11 years ago. For some reason while watching the odometer of a vehicle he started adding the digits of the multiplications of the number eight until one digit remained. It is a recursive algorithm. For example:

8 x 1 = 8

8 x 2 = 16, 1 + 6 = 7

8 x 3 = 24, 2 + 4 = 6

8 x 4 = 32, 3 + 2 = 5

8 x 5 = 40, 4 + 0 = 4

8 x 6 = 48, 4 + 8 = 12, 1 + 2 = 3

8 x 7 = 56, 5 + 6 = 11, 1 + 1 = 2

8 x 8 = 64, 6 + 4 = 10, 1 + 0 = 1

8 x 9 = 72, 7 + 2 = 9

From here the pattern repeats. If the algorithm is repeated over the multiplication table for positive integers the following results:

```
     1  2  3  4  5  6  7  8  9
x    -------------------
1   |1  2  3  4  5  6  7  8  9
2   |2  4  6  8  1  3  5  7  9
3   |3  6  9  3  6  9  3  6  9
4   |4  8  3  7  2  6  1  5  9
5   |5  1  6  2  7  3  8  4  9
6   |6  3  9  6  3  9  6  3  9
7   |7  5  3  1  8  6  4  2  9
8   |8  7  6  5  4  3  2  1  9
9   |9  9  9  9  9  9  9  9  9
```

It is easy enough to continue the multiplication table down and to the right and see that it starts to repeat after the rows and columns of nines, forming a tiling. My friend noticed some other interesting properties of this square.

If you go across any row or column and add the rows or column numbers that add to nine, the sums of the entries of those rows and columns add to nine or "digitally reduce" to nine. For example going across row 2:

The 1^{st} + 8^{th} column = 2 + 7 = 9

The 2^{nd} + 7^{th} column = 4 + 5 = 9

The 3^{rd} + 6^{th} column = 6 + 3 = 9

The 4^{th} + 5^{th} column = 8 +1 = 9

Or across row 3:

The 1^{st} + 8^{th} column = 3 + 6 = 9

The 2^{nd} + 7^{th} column = 6 + 3 = 9

The 3^{rd} + 6^{th} column = 9 + 9 = 18 = 1 + 8 = 9

The 4^{th} + 5^{th} column = 3 +6 = 9

The reader can verify these results for any row or column.

Also if you sum up all nine elements of any row or column and then digitally reduce the value is 9. For example: 1 + 2 +3 + 4 + 5 + 6 + 7 + 8 + 9 = 45 =4 + 5 = 9.

2 + 4 + 6 + 8 + 1 + 3 + 5 + 7 + 9 = 45 = 4 + 5 = 9.

Another interesting property of the square is the unique symmetry it possesses. Disregarding the rows and columns of nines and dividing the square in halves vertically and horizontally:

1).

```
1 2 3 4 5 6 7 8
2 4 6 8 1 3 5 7
3 6 9 3 6 9 3 6
4 8 3 7 2 6 1 5
-----------------
5 1 6 2 7 3 8 4
6 3 9 6 3 9 6 3
7 5 3 1 8 6 4 2
8 7 6 5 4 3 2 1
```

2).

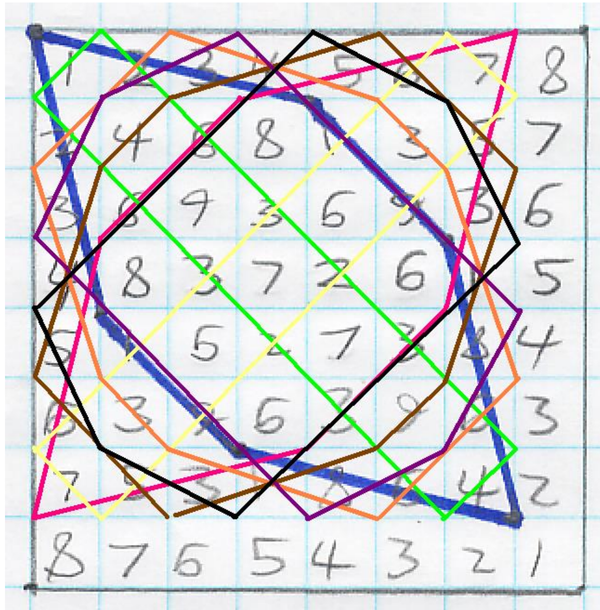
```
1 2 3 4 |5 6 7 8
2 4 6 8 |1 3 5 7
3 6 9 3 |6 9 3 6
4 8 3 7 |2 6 1 5
5 1 6 2 |7 3 8 4
6 3 9 6 |3 9 6 3
7 5 3 1 |8 6 4 2
8 7 6 5 |4 3 2 1
```

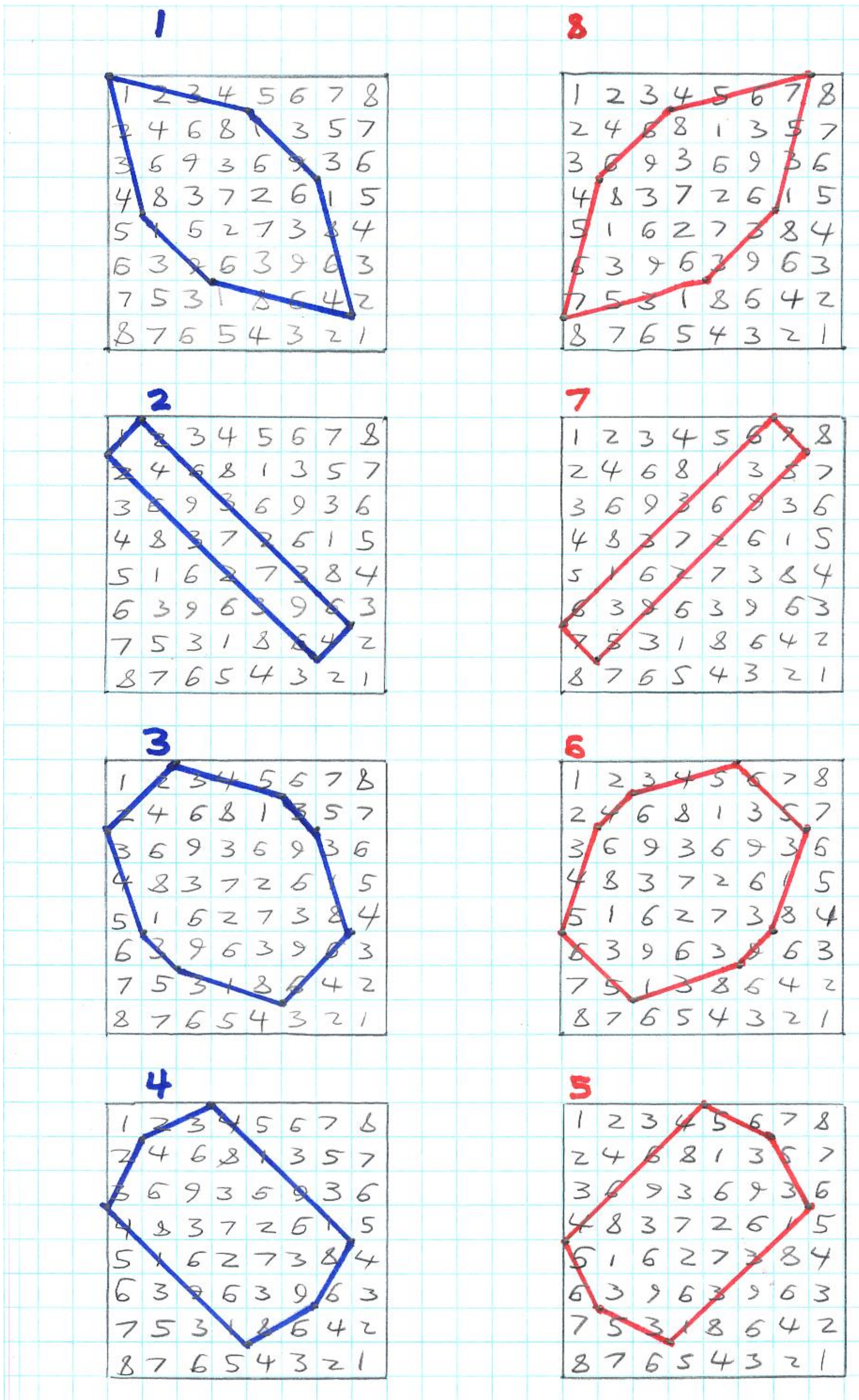
3).

```
1 2 3 4 |5 6 7 8
2 4 6 8 |1 3 5 7
3 6 9 3 |6 9 3 6
4 8 3 7 |2 6 1 5
-------------------
5 1 6 2 |7 3 8 4
6 3 9 6 |3 9 6 3
7 5 3 1 |8 6 4 2
8 7 6 5 |4 3 2 1
```

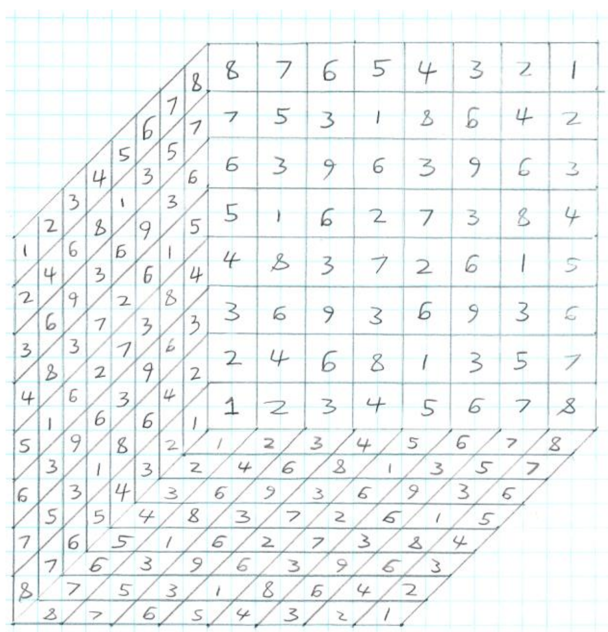
Notice the following properties. The 8th row is backwards from the 1st row, and both are the same as the 8th and 1st column. Likewise for the 7th and 2nd , 6th and 3rd, and 5th and 4th rows and columns. These all add to nine. Everything that goes down on the left side goes up on its reflection on the right side. Everything that goes left to right on the top goes right to left on the bottom. The top half is the bottom half flipped over. The left half is the right half flipped over. And the third figure illustrates how the top left corner is upside down and backwards from the bottom left corner, and the same for the top right and bottom left corners.

One property that many people found in this square are the special geometries each digit possesses. If the square was made of wood and you put nails in all the 4's and then snapped a rubber a band around them, they'd form a polygon. If you did the same for all the 5's, it would form the same polygon but flipped (upside down or sideways). Likewise for all the pairs that add too 9 (1 and 8, 2 and 7, 3 and 6). One might begin to see that each digit possesses a specific "shape" under digital root base 10, a polygon associated with the number (and the number base as we shall see). The following figures illustrate:




1
2
3
4
5
6
7
8

Another very interesting property of this matrix my friend found is this. Notice that regardless of the direction you go, if you start at an edge number the sequence is always the same (nines not included, they are the "shell"). If you start at a one the sequence is 1 2 3 4 5 6 7 8. If you start at a 2 the sequence is 2 4 6 8 1 3 5 7, and so on. The nines form the divider between squares. Now if you can, imagine this figure flat on the plane. Everywhere there is a 1 start filling in the 1 sequence going up into a 3 dimensional grid, likewise for 2, 3, 4, etc. The top row obviously tiles an identical but mirror copy of the original up into 3 dimensions, a "back wall". It will not take you long to see that this forms a cube with nines as its "shell", except for the "zeros" along the x, y, and z planes which we aren't in the set of positive integers. The front of the cube is the back flipped around, the top is the bottom flipped over, and the sides are the same but turned around. The 1st horizontal layer is the 8th layer flipped upside down. Likewise for the 2nd and 7th layer, 3rd and 6th layer, etc. This also applies to vertical layers both from side to side and front to back.



Also, see the slices one by one in the math section. In the mathematical analysis, I will show how I deduced the simple addition to the function that creates the 3-d (or n-dimensional) cube.

These are the main properties my friend found, plus a few others I will not go into. So for 11 years he kept it to himself, showing only friends but never revealing how he created it. Finally last year, knowing he wanted me to check out his work, I asked him to divulge his secret algorithm for creating this figure. He acquiesced, and with the algorithm I set about exploring all these observations mathematically. On the recommendation of yet another friend, David Skelding, I tried the algorithm in other number bases, and lo and behold they produced different cubes exhibiting similar properties. Finally, on the recommendation of yet a third person, I decided to risk the “secret” and entered a few rows of the matrix into Google in vector form. Since the first row is “123456789” the first 3 rows yielded millions of results, none of them relevant. Refining the search I tried “24681357 : 369369369”, the second two rows. This hit pay dirt: One result, a math forum, and there was the matrix. Of course that site has now mysteriously disappeared and I can no longer find it, but it showed me that other people had found this symmetric matrix, and its name, the "Vedic Square".

It turns out the process of recursively adding the digits of a number until one digit remains is called the digital root, and this magic square has origins in ancient India, where the Indian-Arabic number system was created. The history is disputed and unreliable in some cases, and many sources and web pages have no references, leaving it still quite mysterious. By searching for the same string with spaces between the numbers, the term "Vedic Square", and "digital root", I found much more information on dozens of web sites. Some web pages are postings of people who, like my friend Kaya, found it on their own without ever seeing it before. Some of the web sites are K-12 school sites where the square and its patterns are being used to inspire children about math and the multiplication table.

On many web pages about this magic square people noted that it appears to be identical to multiplication mod 9, except with 9's everywhere there are 0's in the modulus. This is the critical fact that allows (easy) in depth mathematical analysis, so I have written a proof of the congruence of the digital root to the modulus. Before I go into the mathematics, applications, and how I took the next step into new algorithms and applications, I will give a very brief history of the number 9, the "vedic square" and "vedic" mathematics.

History

To start off, all I can say is that "vedic" mathematics is a subject of much dispute, and that (according to many varied sources) most religious leaders in India insist that there are no hidden numerical formulas in the Sutras of the Vedas, and that this is a modern hoax combined with a bunch of old mathematical tricks known by people in India. And many mathematicians will point out that the techniques do not work in every case or have to be changed case by case, and therefore are not a complete mathematics, like our tedious Western long multiplication and division are. The square itself seems to be known well for a long time in India. University libraries are not particularly rich on the subject, so I will have to get most of my references from the Internet

According to Harish Johari,

> "Nothing about the origin of the Vedic Square is known in India. The designs obtained from this square are used as decorative patterns in various palaces and shrines throughout India... Muslim artists and craftsmen as well have used the Vedic Square to obtain many of their patterns." (Johari 18)

It seems that nine is the "magic number" of this square, and as well in Vedic Mathematics in general. The digitals roots of large numbers can be taken quickly by "casting out nines", and nine is important in most of the observations. Most of the Vedic methods are based on mod 9 properties, and its importance in our base ten number system as the "roll over" number cannot be understated. Here I leave the reader to do their own exploration, and begin the "real" math and steer cautiously away from the shores of numerology.

Mathematical Analysis

The first thing I had to try was other number bases, on the recommendation of a second friend. Now to take a digital root properly in other bases arithmetic must be performed properly in that base. I tried a few bases and got similar patterns and symmetry for any number base I tried:

Base 8	**Base 9**	**Base 6**	**Base 7**
1 2 3 4 5 6 7	1 2 3 4 5 6 7 8	1 2 3 4 5	1 2 3 4 5 6
2 4 6 1 3 5 7	2 4 6 8 2 4 6 8	2 4 1 3 5	2 4 6 2 4 6
3 6 2 5 1 4 7	3 6 1 4 7 2 5 8	3 1 4 2 5	3 6 3 6 3 6
4 1 5 2 6 3 7	4 8 4 8 4 8 4 8	4 3 2 1 5	4 2 6 4 2 6
5 3 1 6 4 2 7	5 2 7 4 1 6 3 8	5 5 5 5 5	5 4 3 2 1 6
6 5 4 3 2 1 7	6 4 2 8 6 4 2 8		6 6 6 6 6 6
7 7 7 7 7 7 7	7 6 5 4 3 2 1 8		
	8 8 8 8 8 8 8 8		

Notice that even bases have an odd "magic number", namely base-1, and therefore an even number of entries in their square (base-2, not including the "shell"), which split evenly into vertical and horizontal halves. These have 180° rotational symmetry. Notice that odd bases have even "magic numbers", and therefore the square has an odd number of elements. Notice these odd squares form a "divider row and column" that spin 180° and leave an even number of elements on both sides to mirror with, also through 180°.

Like any good mathematician, I abhor tedious calculation. So I devised a way to program the recursive algorithm in Matlab. Here is the code:

```
function A = kayas_square(base)
% version 1 integers 1- (base-1) only

for i = 1:base-1
    for j = 1:base-1
        temp = i * j;
        while temp>(base-1)
            temp = floor(temp/base)+ mod(temp,base);
        end
        A(i,j) = temp;
    end
end
```

Here's how this works. The loops go from 1 to base-1, the magic number. Then temp holds the multiplication of the row and column numbers. The next “while” statement is the recursion. It loops until the number is digitally reduced to the magic number or lower. Taking the floor of (number/base) picks off the right most digit in that base and leaves everything to the left of it, a zero if there are no left digits. The mod in that base yields the right digit. Then the digits are added together. If the resulting number is still larger than base-1 the algorithm repeats and the number reduces again. One nice thing about the digital root is that you can add the digits up any way you want. For example the digital root base 10 of 244 is: 2 + 4 + 4 = 10 = 1+0 = 1. But you could just have well added 24 + 4 = 28 = 2 + 8= 10 = 1. This is why the computer code always works for any size number. The upcoming proof addresses this, as I need to justify this statement. So lets look at some more results:

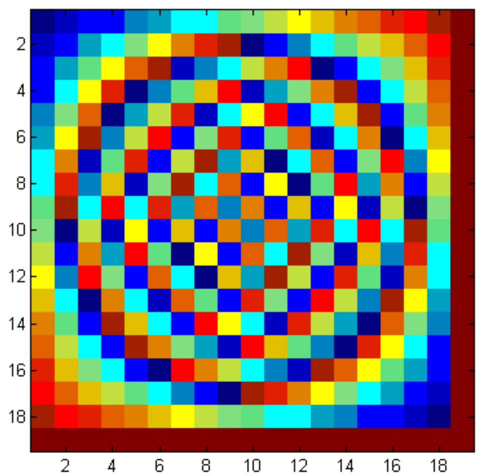
```
EDU>> kayas_square(11)
ans =

     1     2     3     4     5     6     7     8     9    10
     2     4     6     8    10     2     4     6     8    10
     3     6     9     2     5     8     1     4     7    10
     4     8     2     6    10     4     8     2     6    10
     5    10     5    10     5    10     5    10     5    10
     6     2     8     4    10     6     2     8     4    10
     7     4     1     8     5     2     9     6     3    10
     8     6     4     2    10     8     6     4     2    10
     9     8     7     6     5     4     3     2     1    10
    10    10    10    10    10    10    10    10    10    10

EDU>> kayas_square(12)
ans =
     1     2     3     4     5     6     7     8     9    10    11
     2     4     6     8    10     1     3     5     7     9    11
     3     6     9     1     4     7    10     2     5     8    11
     4     8     1     5     9     2     6    10     3     7    11
     5    10     4     9     3     8     2     7     1     6    11
     6     1     7     2     8     3     9     4    10     5    11
     7     3    10     6     2     9     5     1     8     4    11
     8     5     2    10     7     4     1     9     6     3    11
     9     7     5     3     1    10     8     6     4     2    11
    10     9     8     7     6     5     4     3     2     1    11
    11    11    11    11    11    11    11    11    11    11    11
```

At this point it was obviously going to get hard fit on the screen as text, so I decided to map the matrix as a color scaled bitmap instead. Matlab has a built in command "imagesc(Matrix A)" that does this. Lets look further:

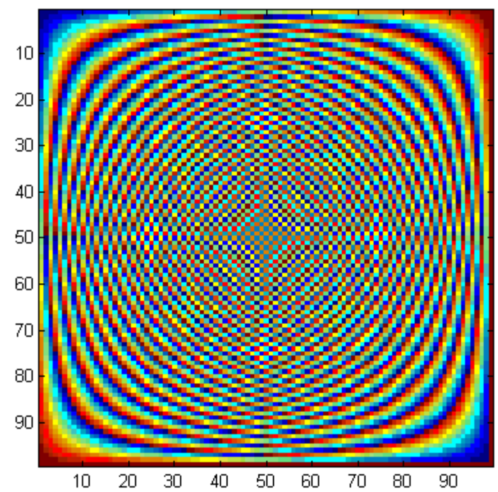
```
EDU>> A= kayas_square(20);
```

EDU>> imagesc(A);

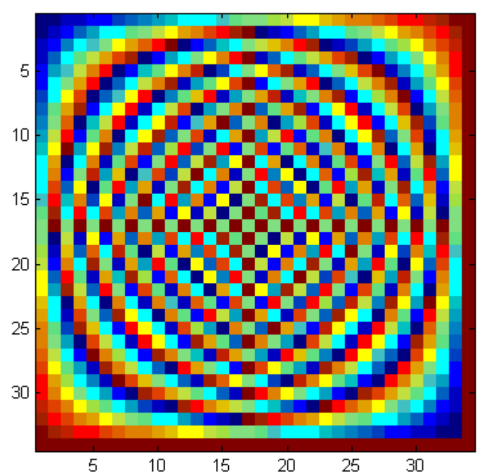
Base 20:



Base 100:



Base 35:

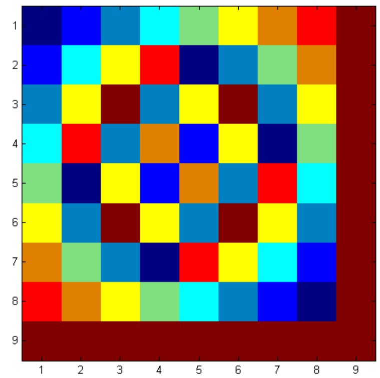


The same 180° rotational symmetry is always there. At this point I wanted to figure out how the 3-d cube worked. Naturally I thought to multiply by z since the x and y coordinates were being multiplied. Graphing a 3-d matrix is tough, it's "solid", it has 3 dimensions and a value at each 3-d location, and requires transparency and markers to graph. I have yet to figure out the best scheme of doing that, but it will probably look like 3-d checkers. So, for now I have been looking at one z layer at a time to analyze. This program has a second parameter z that defines the z coordinate, and it draws a 2-d layer at that height:

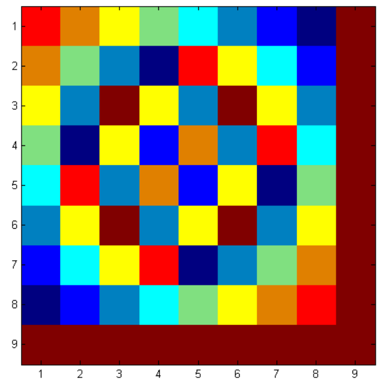
```
function A = kayas_square_z(base,z)
% version 1 integers 1- (base-1) only
for i = 1:base-1
    for j = 1:base-1
        temp = i * j *z;
        while temp>(base-1)
            temp = floor(temp/base)+
mod(temp,base);
        end
        A(i,j) = temp;
    end
end
```

Now lets look at the base ten cube comparing slices that add up to nine:

z=1

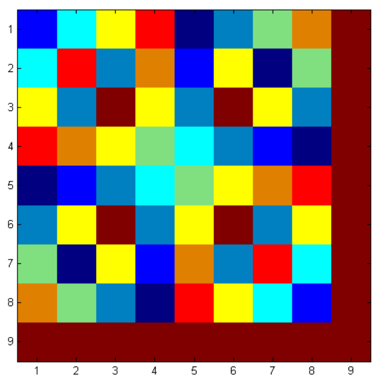


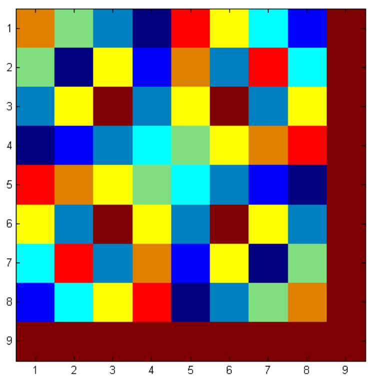
z=8



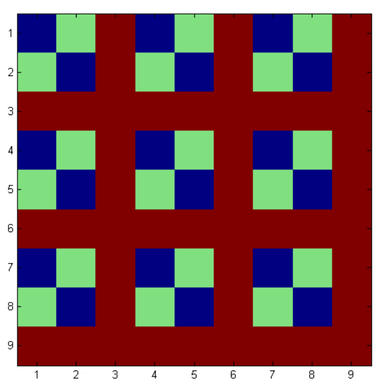
z=2

z=7

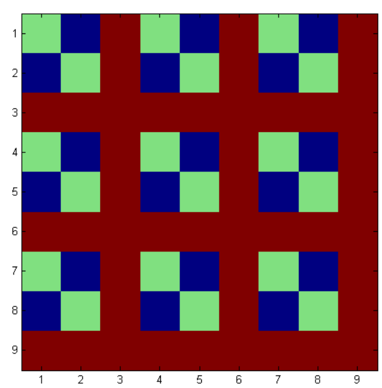




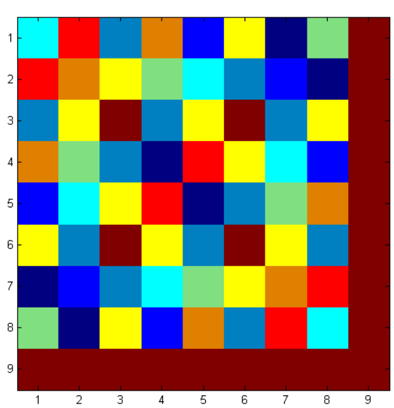
z=3



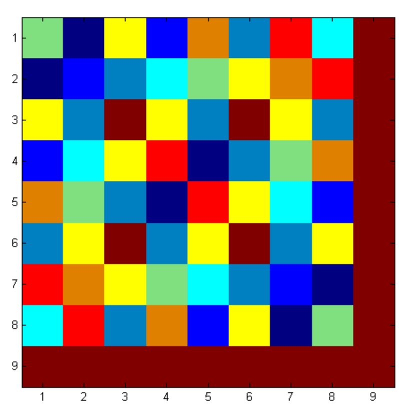
z=6



z=4



z=5

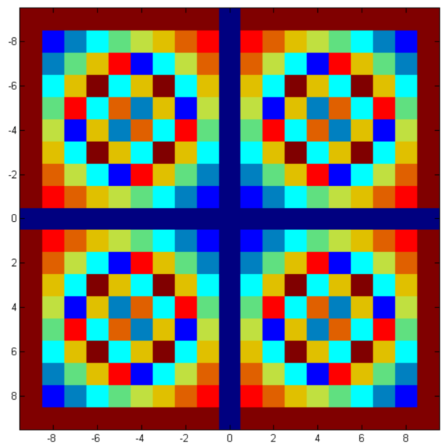


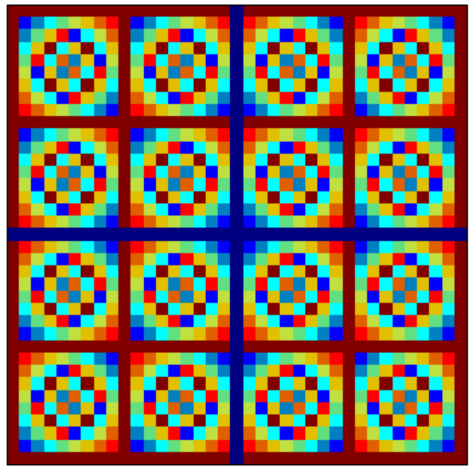
The 1st and 8th, 2nd and 7th, 3rd and 6th, and 4th and 5th layers are "flips" of each other. By mentally swapping the y and z or the x and z axes you can see the slices also go front to back and side to side in the cube. At z = 9 a solid plane of 9's exists, followed by a repeat of the cube, another plane of 9's at z=18, and so on. Now a few questions pop up. What are the values of this function in other quadrants? And what are the values between integers, if we dare take the digital root of non-integers?

First, as written, this doesn't work in any quadrant in which multiplication is negative. The algorithm just returns x*y*z because the value is always less than the magic number and the recursion never executes. If you redefine the algorithm as the digital root of the absolute value of x*y*z it reflects across the origin into all 8 eight octants of 3-d space, which is simple to see. Here I modified the code to run from -magic# to +magic#, using the absolute value. This covers the four quadrants around the origin, and shows the 4 way reflection about the origin.

```
function A = double_square(base,z)
% version 2 integers from -base+1 to +base-1
% uses absolute val before digital root
xIndex= 1;
yIndex= 1;
for i = -base+1:base-1
    for j = -base+1:base-1
        temp = abs(i * j *z);
        while temp>(base-1)
            temp = floor(temp/base)+
mod(temp,base);
        end
        A(xIndex,yIndex) = temp;
        yIndex = yIndex +1;
    end
    xIndex = xIndex+1;
    yIndex = 1;
end
```

Here is what the digital root base 10likes tiled 4 ways and 16 ways about the origin:



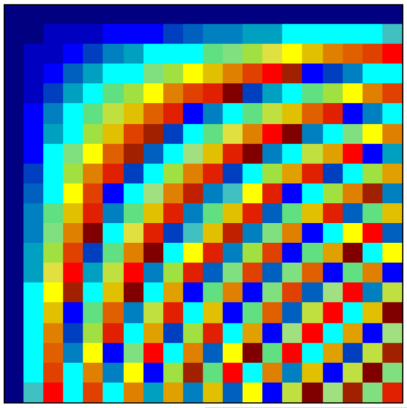


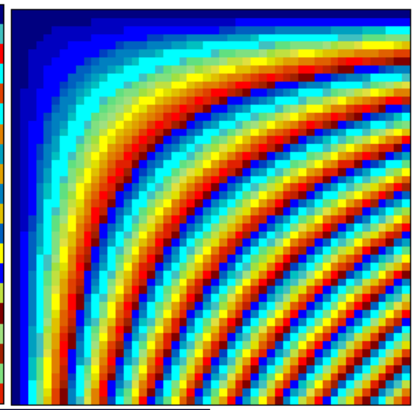
Ok that is one mystery solved. In negative z quadrants it looks just the same.

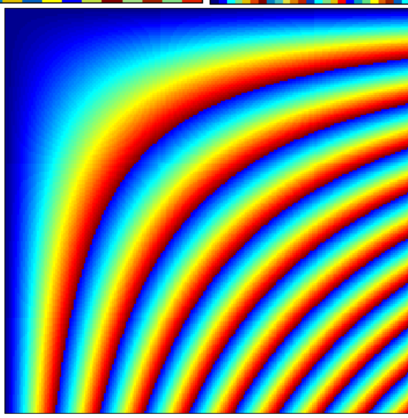
So what about in between integers? I went ahead and ran the program over non-integer values in between the integers because nothing prevented me from doing so. I just replaced the mod with the remainder function. Then I divide the interval from 0 to the magic number into any number of specified subintervals. Here's the code:

```
 function A = fractional_square(base,size,z)
%goes on non-integer values
%size divisions between 0 and base-1
A = zeros( size);
for i = 0:size-1;
    for j = 0:size-1;
        temp = i*(base-1)/(size-1)*j*(base-
1)/(size-1)*z;
        while temp>=base
            temp = floor(temp/base)+
rem(temp,base);
        end
        A(i+1,j+1)  = temp;
    end
end
```

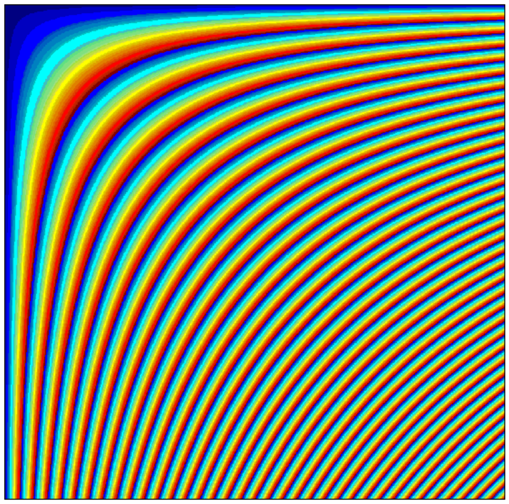
Sampled at 20, 50, and then 200 subintervals, with 0 now included in the range:







Where's the Vedic Square? Here is a smooth continuous function with jump discontinuities everywhere the digital root "rolls over" at nine (where red goes to blue). This underlies the integer matrix we were examining. As long as we observe the function at discrete values and an even spacing we see a different pattern, at the integers it's the Vedic Square. The level curves get closer as you get further from the origin. The equations for the level curves where the digital root base 10 = 9 are: xy= 9, 18, 27, 36, ... etc., or y = 9/x, 18/x, 27/x,... etc. These are hyperbolas. The distance between the level curves is proportional to 1/x^2 (or 1/y^2) along the line y=x (The main diagonal of the matrix). This figure shows the digital root base 10 out to x=18, y=18:



The patterns we saw are the result of laying a square grid on this and sampling the grid points (4 Vedic Squares lie on this picture). And yet the pattern at the integers repeats the same Vedic Square out to infinity. So how can I make this claim? Here is where the congruence to the modulus comes into play.

Suppose we label a number in a number base : $d_n d_{n-1} d_{n-2} .... d_2 d_1 d_0$. As each digit is removed from the base^n column and moved to the ones column (base^0) we are changing the number by a value ($-d*b^n + d*1$) or $-d(b^n-1)$. The net change in the value of the number in the first step of digitally reducing a number with k digits=

$$\sum_{n=0}^{n=k-1} -d_n (b^n - 1)$$

If this sum is <= base-1 we stop, otherwise run again. In either case the net change will be the sum of the ($-d_n$)'s times the ($b\wedge^n-1$)'s from all the recursive steps. What I want to prove is that each term in the sum is congruent to 0 mod (b-1), i.e. that any integer $k*(b^{\wedge n}-1)$ is congruent to 0 mod(b-1).

Symbolically, prove:

[$(b^{\wedge n}-1) \equiv 0 \bmod (b-1)$ for any integers n, b >1], thus proving it for all the $(b^{\wedge n}-1)$'s in our summation. The obvious tool of choice here is Mathematical Induction. By proving a formula for a basis case and proving that I can get from any arbitrary nth case to the nth + 1 case I prove an infinite line of cases, like a line of dominoes falling. So here goes. There is a fork in the proof where the induction is proven two ways (there are more ways to prove this also):

first I need a couple of Lemmas:

Lemma a):

1). if $a\equiv 0$ mod(m) then a=sm for some integer s (thus the 0 remainder when dividing by m)

2). if $b\equiv 0$ mod(m) then b=tm for some integer t

3). therefore a + b = sm + tm = (s+t)m = km where k is an integer and thus:

4). $a+b\equiv 0$ mod(m) because the remainder of km/m is zero therefore:

5). If $a\equiv 0$ mod(m) and $b\equiv 0$ mod(m) then $(a+b) \equiv 0$ mod(m)

Lemma b):

1). if $a\equiv 0$ mod(m) then a =sm for some integer s

2). Therefore ka = k(sm)=(ks)m= Km for any integer k, and K is another integer.

3). Km a≡0 mod(m) because Km/m has a zero remainder

4). Therefore if a≡0 mod(m) then ka≡0 mod(m) for any

integer k

Proceeding with the proof:

Basis case:

n=2

b^2-1 = (b+1)(b-1)= k(b-1), integer k,

therefore (b^2-1) ≡ 0 mod(b-1) (via Lemma b).

Induction Step:

Suppose (b^n-1) ≡ 0 mod(b-1) for some arbitrary n:

1). (b^n-1) ≡ 0 mod(b-1)

FORK I:

2). (b-1)b^n ≡ 0 mod(b-1) ((b-1)≡0 mod(b-1) + Lemma a).)

3). [(b^n-1)+ (b-1)b^n] ≡ 0 mod(b-1) (Lemma a) and 1). and 2).)

4). [b^n- 1 + b^n(b-1)] ≡ 0 mod(b-1)

5). [b^n + b^n(b-1) -1] ≡ 0 mod(b-1)

6). [b^n (1+ (b-1)) -1] ≡ 0 mod(b-1)

7). [b^n (1+ b -1) -1] ≡ 0 mod(b-1)

8). [b^n (b) -1] ≡ 0 mod(b-1)

9). [b^{n+1} -1] ≡ 0 mod(b-1)

Thus by the I.H. we have proved (b^^{n}-1) ≡ 0 mod(b-1) for all integers

b,n>1

FORK II:

1). $(b^n-1) \equiv 0 \bmod(b-1)$ (reiterated)

2). $b(b^n-1) \equiv 0 \bmod(b-1)$ (Lemma b).)

3). $b(b^n-1) + (b-1)\ ) \equiv 0 \bmod(b-1)$ (Lemma a). applied to 1). and 2).)

4). $b^{n+1}-b + b -1 \equiv 0 \bmod(b-1)$

5). $[\ b^{n+1} -1] \equiv 0 \bmod(b-1)$

Thus by the I.H. we have proved $(b^n-1) \equiv 0 \bmod(b-1)$ for all integers b, n>1

Let me argue that we are done here. The net change in the number was the sum of the $(k*(b^n-1))$'s. But we proven all the (b^n-1)'s are congruent to 0 mod (b-1). In effect we have cast out (b-1)'s, and are left with a number between 1 and b-1. This must be the either b-1 or the remainder after division by b-1. Thus, the digital root of x base m = x mod(m-1), unless x mod(m-1) = 0, in which case it returns m-1. For example, the digital root of x base 10 = x mod (9) if x mod (9) $\neq$ 0; 9 if x mod(9) = 0. We just substitute 9 every there is a zero in the mod. That is just a matter of semantics. It is the same function for all practical purposes, although the algorithm is unconventional. That is good, because far more theorems and proofs are written on the properties of the modulus than the digital root. And writing proofs for a recursive algorithm could prove to be very difficult, as the previous proof illustrates. The modulus is also way faster to compute and easier to work with. Let's see our Vedic Square as a mod 9 square, with the code (now much simpler):

```
function A = kayas_square_mod(modulus)
% USING THE MODULUS INSTEAD OF THE DIGITAL ROOT
% Goes from 0 to the modulus

for i = 0:modulus
    for j = 0:modulus
        temp = i * j;
        A(i+1,j+1) = mod(temp,modulus);
    end
end

EDU>> A=kayas_square_mod(9);
EDU>> A

A =

     0    0    0    0    0    0    0    0    0    0
     0    1    2    3    4    5    6    7    8    0
     0    2    4    6    8    1    3    5    7    0
     0    3    6    0    3    6    0    3    6    0
     0    4    8    3    7    2    6    1    5    0
     0    5    1    6    2    7    3    8    4    0
     0    6    3    0    6    3    0    6    3    0
     0    7    5    3    1    8    6    4    2    0
     0    8    7    6    5    4    3    2    1    0
     0    0    0    0    0    0    0    0    0    0
```

And there it is. The mod 9 square. The nines in the Vedic Square are replaced by zeros.

So let's prove some of my friend Kaya's observations have a mathematical basis. Lets assume a general case of a m*m mod(m) square. First lets prove that the sums of the 1st and (m-1)st, 2nd and (m-2)th entries, etc., of each row and column are congruent to 0 mod(m) . Label the matrix $A_{i,j}$ where i, j are the rows and columns and go from 1 to m where m is the modulus. Some immediate observations:

1). This is a symmetric matrix in classical matrix terms. What this means is reflection across the main diagonal, i.e. $A_{i,j}$= (i*j) mod m=(j*i) mod m= $A_{j,i}$. Also, this means A = A^T (A transpose: switching the rows and columns). This is due to the commutative property of multiplication.

2). Because the matrix is symmetric anything proved for the rows is proved for the columns. I could easily switch the axes labels and it wouldn't matter. It would just the transpose of the matrix.

So I will just prove this for the rows and it will apply to the columns as well.

We have a modulus square mod(m) Let's take the two opposing entries on some row r on columns k and (m-k). The sum of these entries is rk mod(m) + r(m-k)mod(m).

There is a theorem in modular arithmetic that states:

"Theorem 10: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$. " (Rosen 163)

Let $a = rk$

Let $b= rk \bmod(m)$

$a \equiv b \pmod{m}$

Let $c = r(m-k)$

Let $d= r(m-k) \bmod m$

$c \equiv d \pmod{m}$

$a + c \equiv b + d \pmod{m}$ (from theorem)

$rk+ r(m-k) \equiv c + d \pmod{m}$

$r(k + (m-k)) \equiv c + d \pmod{m}$

$rm \equiv c + d \pmod{m}$

$rm \equiv 0 \pmod{m}$ therefore $c + d \equiv 0 \pmod{m}$. But c + d is our sum, and this is what we wanted to prove.

Let's look at why the sums of all the columns or rows are congruent to 0 (mod m). Let's look at the first column or row only. In that column or row is simply the summation 1 + 2 + 3 ... + (m-2) + (m-1) + m = $\sum_{k=1}^{k=m} k$. I want to prove $\sum_{k=1}^{k=m} k \equiv 0$ (mod m). Let's take a look at the sums of this arithmetic series:

1 = 1	1 ≡ 0 (mod 1)	= 1 * 1
1 + 2 = 3	3 !≡ 0 (mod 2)	= 1 * 3
1 + 2 + 3 = 6	6 ≡ 0 (mod 3)	= 3 * 2
1 + 2 + 3 + 4 = 10	10 !≡ 0 (mod 4)	= 2 * 5
1 + 2 + 3 + 4 + 5 = 15	15 ≡ 0 (mod 5)	= 5 * 3
1 + 2 + 3 + 4 + 5 + 6 = 21	21 !≡ 0 (mod 6)	= 3 * 7
1 + 2 + 3 + 4 + 5 + 6 + 7 = 28	28 ≡ 0 (mod 7)	= 7 * 4
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 = 36	36 !≡ 0 (mod 8)	= 4 * 9
1+2+3+4+5+6+7+8+9=45	45 ≡ 0 (mod 9)	= 9 * 5

Look at the products on the far right. Apparently, for m odd $\sum_{k=1}^{k=m} k$ = m((m+1)/2), and for m even $\sum_{k=1}^{k=m} k$ = m/2(m + 1). But m((m+1)/2)= m/2(m + 1), so it is the same formula, algebraically. So the next step is to prove this observation, and since this is a series I will use Induction again.

Basis case, m=1:

$$\sum_{k=1}^{k=1} k = 1 = 1\ ((1+1)/2)$$

Induction Step:

Suppose $\sum_{k=1}^{k=m} k = m\ (\ (m+1)/2\ )$ for some arbitrary m.

now if we add (m+1) to both sides:

$$\sum_{k=1}^{k=m} k\ + (m+1) = m\ (\ (m+1)/2\ ) + (m+1)$$

$$\sum_{k=1}^{k=m+1} k\ = m/2(m+1) + (m+1)$$

$$\sum_{k=1}^{k=m+1} k\ = m^2/2 + m/2 + m + 1$$

$$\sum_{k=1}^{k=m+1} k\ = m^2/2 + 3m/2 + 1$$

$$\sum_{k=1}^{k=m+1} k\ = 1/2(m+1)(m+2)$$

Therefore, by the I.H we have proved that $\sum_{k=1}^{k=m} k = m\ (\ (m+1)/2\ )$ for m = 1,2,

3, 4, 5, 6, 7....

If m is odd then (m+1)/2 is and integer, if m is even (m+1)/2 is not an integer. In the odd case we have proved that the sum of the first row is equal to some integer k * m, therefore is congruent to 0 (mod m) which is what I wanted to prove. If m is even then the sum of the first row is equal to some non-integer r * m, therefore is not congruent to 0 (mod m). Hence:

1). If a positive integer m is odd then $\sum_{k=1}^{k=m} k \equiv 0 \pmod{m}$

Now I need to extend the Rosen proof of Theorem 10 (Rosen 163) to apply to a sum of a series of numbers instead of just two numbers:

First I need another theorem:

"Theorem 9: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that a = b + km. " (Rosen 162).

Proceeding:

Let a' = a mod m, b' = b mod m, c' = c mod m,....

Therefore a ≡ a' (mod m), b ≡ b' (mod m), c ≡ c' (mod m)....

If a ≡ a' (mod m) and b ≡ b' (mod m) and c ≡ c' (mod m)....

then there are integers s, t, u, ... with a' = a + sm, b' = b + tm, c' = c + um....

By Theorem 9:

a' + b' + c' +... = (a + sm) + (b+ tm) + (c + um)+....

a' + b' + c' +... = (a + b + c +....) + (sm + tm + um....)

a' + b' + c' +... = (a + b + c +....) + (s + t + u +....)m

(a' + b' + c' + ...) = (a + b + c +....) + km, with integer k

And by theorem 9 and from line above,

(a' + b' + c' + ...) ≡ (a + b + c + ...) (mod m)

Lemma c).

If a' = a mod m, b' = b mod m, c' = c mod m,....

then (a + b + c + ...) ≡ (a' + b' + c' + ...) (mod m)

Applying Lemma c). to the modulus of the arithmetic series:

$$\sum_{k=1}^{k=m}(k \text{ mod m}) \equiv [\sum_{k=1}^{k=m} k\,] \text{ (mod m)}$$ and furthermore that for any row r:

$$\sum_{k=1}^{k=m}(rk \text{ mod m}) \equiv [\sum_{k=1}^{k=m} rk\,]\text{(mod m)} = [r\sum_{k=1}^{k=m} k\,]\text{(mod m)},$$ r constant integer.

Finally, using the second multiplicative part of the theorem 10 along with result 1). from this proof:

$$r\sum_{k=1}^{k=m} k \text{ (mod m)} \equiv r*0 \text{ (mod m)} \equiv 0 \text{ (mod m) for m odd}$$

and finally:

$$\sum_{k=1}^{k=m}(rk \text{ mod m}) \equiv 0 \text{ (mod m) for m odd}$$

This is why the sums of all the rows and columns of the Vedic Square digitally reduce to nine. This will work for modulus squares with an odd modulus, like 9.

Another property is the 180° rotational symmetry of the Vedic Square. What I want to prove is the top left entries are the bottom right entries reflected across the center point and that the bottom left entries are the top right entries reflected across the center point. Looking at the rows and columns from 1 to (m-1) what we need to prove is $A_{i,j} = A_{m-i,m-j}$ or that:

ij ?≡? (m-i) (m-j) (mod m)

ij ?≡? m^2 -m(j+i) + ij ≡ 0 + 0 + ij (mod m)

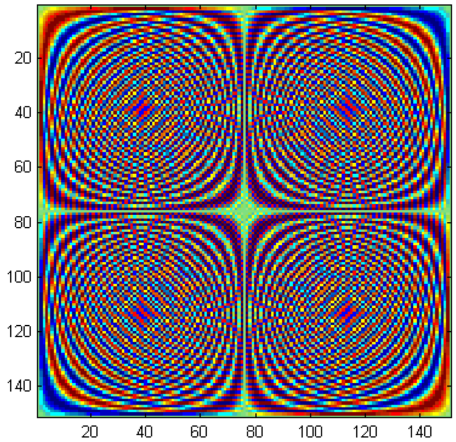
ij ≡ (m-i) (m-j) (mod m)

That concludes the proofs.

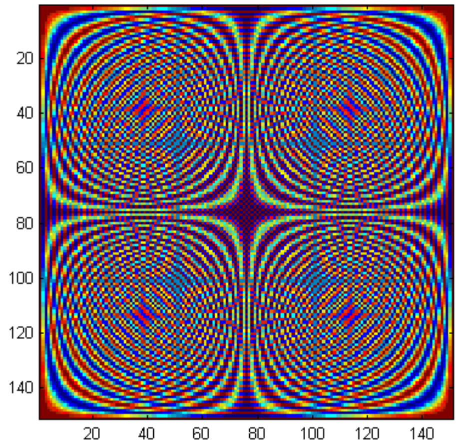
After looking at the symmetry of the Vedic Square, I thought "Any periodic function lined up to terminate at a fixed regular intervals ought to work just like the modulus." I immediately thought of trig functions. Here's the code for a cosine square (sine is the same but with "sin" instead of "cos"):

```
function A = coscube(mod, z)
% creates a cosine cube from 0 to mod
for i = 0:mod
    for j = 0:mod
        A(i+1,j+1) = cos(i*j*z*2*pi/mod);
    end
end
```

Here's a look at the sine and cosine cube at z = 1, size 150:



Sine(x*y*2*pi/150)



Cosine(x*y*2*pi/150)

What do you see in the patterns? What is similar and what is different between sine and cosine? What does this have to do with even and odd functions, and with the orthogonal nature of sine and cosine? I'm going to stop here and let the reader ponder that. The point I wish to make here it is that non-tiling continuous functions can have a repeating symmetric tiling when sampled on discrete integer based grids. Even though trig functions of x*y oscillate faster and faster the further we get from the origin, these discrete integer samplings (adjusted to 2*pi) maintain an even repeating pattern, theoretically out to infinity, disregarding round off error.

Conclusion

All I can say is that I hope I can continue to keep learning more about this phenomenon of these tiling algorithms, and that I hope the reader enjoyed learning about them. I feel that this is a special and unique object. This is the type of thing that rare, creative, experimental, and quirky individuals will probably find time and time again. I believe it is a Universal object, that any intelligent life that discovers written numbers will find these patterns long before they discover calculus. After all, it involves only multiplication and addition and can be done easily by hand. And our hands are our "digital" computer before all other computers.

Works Cited

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